AI001 – Define Data Schema

1. Task Overview

- **Task ID:**
  AI001
- **Task Name:**
  Define Data Schema
- **Task Lead:**
  Preetham Chandu (Integration & Architecture Lead)
- **Team Members:**
  Danny Zhao (Hazard Data Lead) & Jasmanpreet Singh (Cyber Threat Data Lead)
- **Sprint:**
  Sprint 1
- **Date Started:**
  21/03/2026
- **Date Completed:**
  04/04/2026

2. Objective

- **Goal:**
  To establish the foundation of the entire AI/ML pipeline by creating a unified data schema in accordance with BE003 that integrates hazard data, cyber threat data, and shared attributes into a structured JSON format that is compatible with PostgreSQL.

- **Why it matters to the project:**
  This is a critical blocking task as all downstream AI/ML tasks depend on a consistent schema. A poor schema design will negatively impact data cleaning (AI003), feature engineering (AI004), and model training (AI007 onwards). This schema is to ensure data consistency, interoperability, and compatibility with future AI/ML tasks.

3. Task Type

☐ Coding / Implementation

☐ Research / Analysis

☒ Design / Planning

☒ Documentation

☒ Data-related

Other: _____

4. Tools / Approach

- **Tools / Platforms Used:**
  Visual Studio Code, JSON
- **Languages (if any):**
  Knowledge of Python (for ensuring ML compatibility) & SQL (for schema alignment).
- **Approach / Method:**
  Identify key requirements across fields, provide a draft schema, make changes to

align with BE003 deliverables, provide sample data modelling with JSON, ensure AI/ML compatibility throughout schema design.

5. Planned Steps

- **Step 1:** Identify required fields across hazard, cyber, and shared domains.
- **Step 2:** Define naming conventions and datatypes.
- **Step 3:** Design draft JSON for BE003 team for review.
- **Step 4:** Adjust JSON accordingly from feedback after review.
- **Step 5:** Create sample records for hazard and cyber scenarios.
- **Step 6:** Map schema to database with BE003.
- **Step 7:** Validate schema against ML pipeline requirements.

6. Work Completed

- **Implementation / Code:**
  N/A
- **Research Findings / Insights:**
  After review from BE003 team, it was determined that hazard data always requires a structured geolocation and temporal attributes. Cyber threats may not always have a geolocation attached to them. Several fields suggested in the draft may also be normalized to improve scalability.
- **Design / Structure Created:**
  A unified schema that covers hazard data, cyber threat data, risk assessment, and shared attributes.
  JSON structure that aligns with PostgreSQL.

7. Deliverables

- **Output / Code / Files:**

  Sample JSON records, schema mapping and documentation.
- **Summary of Work:**

  A unified data schema designed to integrate hazard events, cyber threats, and risk assessments into a structured JSON format compatible with backend database. Schema defines key attributes across all domains whole ensuring system functionality and AI/ML processability.
- **Supporting Files (docs, datasets, etc.):**

Hazard Threat JSON & Mapping


```
{
"hazard_event_id": "a1f3c2d4-1111-4a2b-b111-000000000001",
"hazard_type": "bushfire",
"description": "Bushfire affecting regional communities",
"severity_level": "high",
"event_status": "active",
"start_time": "2026-03-21T10:30:00Z",
"end_time": null,
"source_id": "33333333-cccc-4ccc-dddd-000000000001",
"source_ref_event": "BOM-12345",
"geo_location_id": "22222222-bbbb-4bbb-cccc-000000000001",
"created_at": "2026-03-21T11:05:00Z",
"updated_at": "2026-03-21T11:05:00Z"
}
```

JSON Field	Database Table	Database Column	Mapping	Notes
hazard_event_id	HAZARD_EVENT	hazard_event_id	Direct mapping	Primary key
hazard_type	HAZARD_EVENT	hazard_type	Direct mapping	Existing field
description	HAZARD_EVENT	description	Direct mapping	Existing field
severity_level	HAZARD_EVENT	severity_level	Direct mapping	Must be one of low/medium/high/critical
event_status	HAZARD_EVENT	event_status	Direct mapping	Must be one of active, resolved, monitoring
start_time	HAZARD_EVENT	start_time	Direct mapping	Existing timestamp field
end_time	HAZARD_EVENT	end_time	Direct mapping	Nullable existing field
source_id	HAZARD_EVENT	source_id	Direct mapping	Foreign key
source_ref_event	HAZARD_EVENT	source_ref_event	Direct mapping	Existing field
geo_location_id	HAZARD_EVENT	geo_location_id	Direct mapping	FK referencing GEO_LOCATION.geo_location_id

JSON Field	Database Table	Database Column	Mapping	Notes
weather_context.temperature_c	—	—	Not currently in BE003	External weather context
weather_context.wind_speed_kmh	—	—	Not currently in BE003	External weather context
weather_context.rainfall_mm	—	—	Not currently in BE003	External weather context
weather_context.fire_danger_index	—	—	Not currently in BE003	External weather context
created_at	HAZARD_EVENT	created_at	Direct mapping	Existing field
updated_at	HAZARD_EVENT	updated_at	Direct mapping	Existing field

Several JSON fields do not map directly to existing database columns as AI001 schema is designed to support API exchange, external data integration, and future AI/ML processing. BE003 schema is more of a storage design. As a result, some fields are retained as optional schema extensions or derived attributes rather than backend columns.

Cyber Threat JSON & Mapping

{
"threat_id": "b2e4d5f6-1111-4c3d-c111-000000000001",
"threat_type": "phishing",

"title": "Fake emergency relief payment email",
"description": "Email campaign impersonating government disaster relief",
"risk_level": "critical",
"status": "active",
"category": "cyber",
"confidence_score": 0.91,
"detected_at": "2026-03-21T12:10:00Z",
"source_id": "33333333-cccc-4ccc-dddd-000000000010",
"created_at": "2026-03-21T12:20:00Z",
"updated_at": "2026-03-21T12:20:00Z"
}

JSON Field	**Database Table**	**Database Column**	**Mapping**	**Notes**
threat_id	CYBER_THREAT	threat_id	Direct mapping	Primary key
threat_type	CYBER_THREAT	threat_type	Direct mapping	Category of threat
title	CYBER_THREAT	title	Direct mapping	short summary
description	CYBER_THREAT	description	Direct mapping	detailed description
risk_level	CYBER_THREAT	risk_level	Direct mapping	Must contain one of Low/Medium/High/Critical

JSON Field	Database Table	Database Column	Mapping	Notes
status	CYBER_THREAT	status	Direct mapping	Must contain one of Active/Monitoring/Resolved/Archived
category	CYBER_THREAT	category	Direct mapping	General grouping
confidence_score	CYBER_THREAT	confidence_score	Direct mapping	Existing field
detected_at	CYBER_THREAT	detected_at	Direct mapping	Must have valid timestamp
source_id	CYBER_THREAT	source_id	Direct mapping	Foreign key
created_at	CYBER_THREAT	created_at	Direct mapping	Existing field
updated_at	CYBER_THREAT	updated_at	Direct mapping	Existing field

Risk Assessment/Integration JSON & Mapping

{
"integration_event_id": "c3f5e6g7-5555-4d4e-d555-000000000005",

"related_hazard_event_id": "a1f3c2d4-3333-4a2b-b333-000000000003",
"related_threat_id": "b2e4d5f6-1111-4c3d-c111-000000000001",
"correlation_score": 0.84,
"linkage_reason": "Threat merged within same region and timeframe during active bushfire response",
"integration_confidence": 0.88,
"linked_event_type": 2,
"event_status": 1,
"event_time": "2026-03-21T12:00:00Z",
"detected_at": "2026-03-21T12:15:00Z",
"reported_at": "2026-03-21T12:25:00Z",
"created_at": "2026-03-21T12:30:00Z",
"updated_at": "2026-03-21T12:30:00Z"
}

JSON Field	**Database Table**	**Database Column**	**Mapping**	**Notes**
integration_event_id	RISK_ASSESSMENT	integration_event_id	Direct mapping	Primary key
related_hazard_event_id	RISK_ASSESSMENT	related_hazard_event_id	Direct mapping	FK linked to HAZARD table
related_threat_id	RISK_ASSESSMENT	related_threat_id	Direct mapping	FK linked to THREAT table
correlation_score	RISK_ASSESSMENT	correlation_score	Direct mapping	Strength of linkage

linkage_reason	RISK_ASSESSMENT	linkage_reason	Direct mapping	Explanation
integration_confidence	RISK_ASSESSMENT	integration_confidence	Direct mapping	Confidence score of linkage
linked_event_type	RISK_ASSESSMENT	linked_event_type	Lookup/ FK mapping	Value resolved via LINKED_EVENT_TYPE table
event_status	RISK_ASSESSMENT	event_status	Lookup/ FK mapping	Value resolved via EVENT_STATUS table
event_time	RISK_ASSESSMENT	event_time	Direct mapping	Main time of occurrence for integration
detected_at	RISK_ASSESSMENT	detected_at	Direct mapping	linked event detected time
reported_at	RISK_ASSESSMENT	reported_at	Direct mapping	linked event time of report
created_at	RISK_ASSESSMENT	created_at	Direct mapping	Existing field
updated_at	RISK_ASSESSMENT	updated_at	Direct mapping	Existing field

The main tables are supported by several reference and join tables to maintain consistency and data integrity. HAZARD_EVENT.geo_location_id links to GEO_LOCATION, while multi-location hazards are supported through HAZARD_LOCATION. Both HAZARD_EVENT and CYBER_THREAT reference the source via source_id. In the integration layer, RISK_ASSESSMENT.related_hazard_event_id and related_threat_id link to HAZARD_EVENT and CYBER_THREAT. The fields linked_event_type and event_status reference the LINKED_EVENT_TYPE and EVENT_STATUS lookup tables. Optional threat geolocation is handled through the THREAT_LOCATION table linking threats to GEO_LOCATION.

GEO_LOCATION Table

Field Name	Data Type	Description
geo_location_id	UUID	Primary key
country	VARCHAR	Name of country
state_region	VARCHAR	state
local_government_area	VARCHAR	LGA
suburb	VARCHAR	Locality
latitude	DECIMAL	latitude coordinate
longitude	DECIMAL	Longitude coordinate
geo_precision	VARCHAR	Level of precision (country/state/LGA/suburb)

LINKED_EVENT_TYPE

Field Name	Data Type	Description
linked_event_type_id	UUID	PK identifier
description	VARCHAR	defines whether record links hazard only, cyber only, or both

EVENT_STATUS

Field Name	Data Type	Description
event_status_id	uuid	PK identifier
event_status_description	VARCHAR	Current state of integrated event

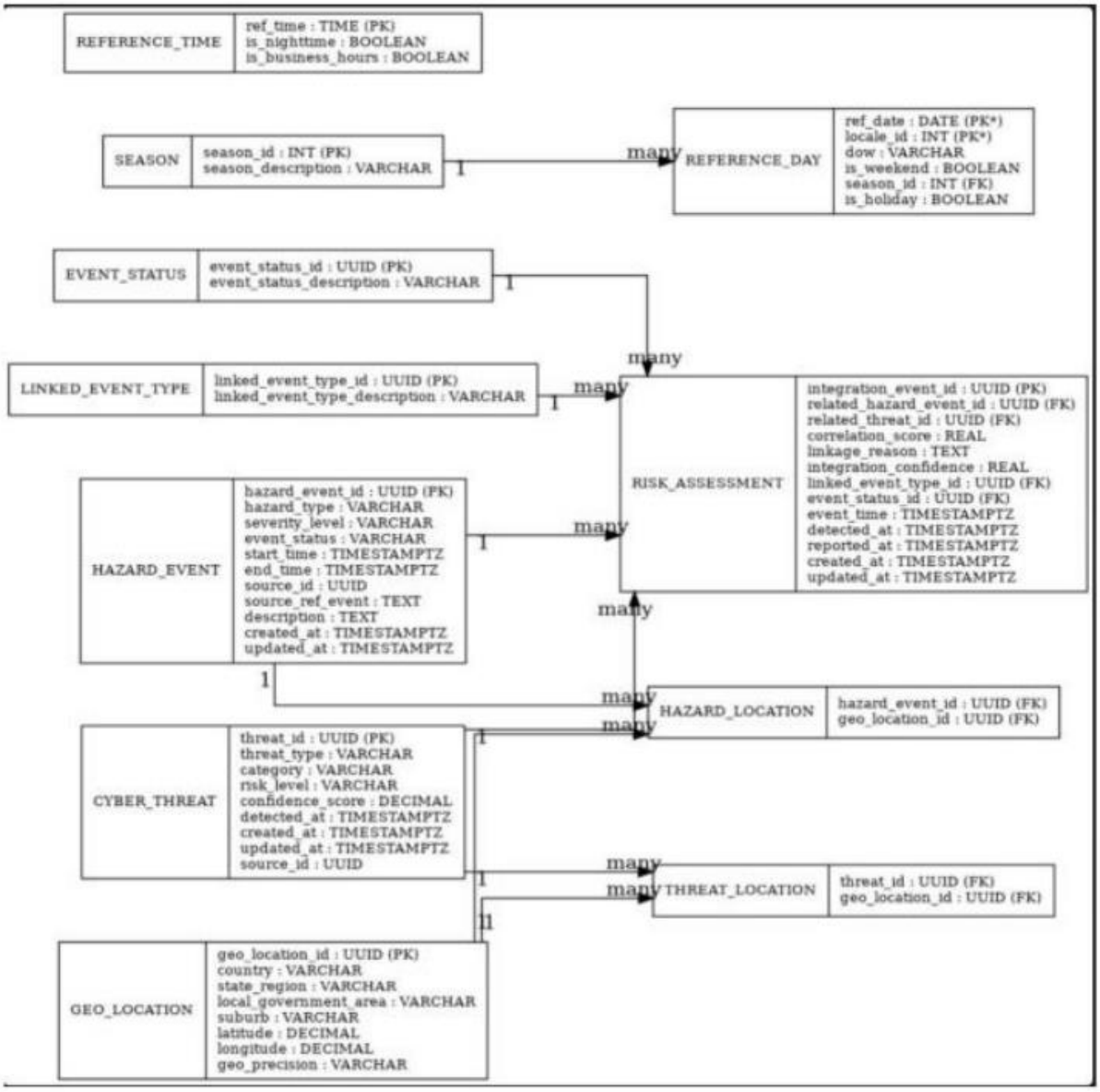
HAZARD_LOCATION (Join Table)

Field Name	Data Type	Description
hazard_event_id	UUID	FK identifier
geo_location_id	UUID	FK identifier

THREAT_LOCATION (Join Table)

Field Name	Data Type	Description
threat_id	UUID	FK identifier
geo_location_id	UUID	FK identifier

Data Relationship



8. Dependencies

- **Depends on:** BE003 as deliverables must align.
- **Blocks / feeds into:** Feeds into AI003 and AI006. Essentially blocks all other AI/ML tasks as this is the first task.

9. Validation / Review

- **How was this tested / verified:**

  This schema was validated through a combination of structural checks, backend team alignment, and AI/ML suitability.

  Alignment with BE003 database schema:

  JSON structures were mapped against the planned backend relational structure to ensure compatibility with tables such as HAZARD_EVENT, CYBER_THREAT, RISK_ASSESSMENT, and GEO_LOCATION. Core fields were verified to have corresponding mappings while nested structures were decomposed into relational tables using foreign keys and join tables.

  Field Validation:

  Each JSON attribute was reviewed to confirm correct datatypes, consistency with backend constraints such as severity levels and status values, and the appropriate use of identifiers.

  Sample record testing:

  Sample JSON records for hazard events cyber threats, and risk assessments were created and manually mapped to the database to ensure all required fields are present, relationships between entities are properly represented and nested structures can be transformed into relational tables.

  Relationship validation:

Relationship validation verified through RISK_ASSESSMENT table, where hazard events and cyber threats are linked. Also ensured many-to-many relationships are correctly handled with HAZARD_LOCATION and THREAT_LOCATION.

AI/ML compatibility check:

Ensured support for downstream AI tasks by consistency in timestamps for temporal analysis, structured geolocation of spatial features, suitable categorical fields for encoding, and placeholders for derived ML metrics.

10. Notes

- **Issues faced:**

  The issues encountered were related to team coordination and alignment of deliverables between the backend (BE003) and AI/ML (AI001) sub-teams. As the project was split into two dependent groups, there was initial uncertainty regarding the scope and structure of each team's deliverables. The overall group formation was also delayed by two days according to AI/ML weekly schedule. This problem persisted again towards the end of Sprint 1, where several Sprint 2 tasks of the backend team were moved to Sprint 1 to ensure no AI/ML blockage, requiring new team formations.

11. Quick Check

☒ **Task is clearly understood**

☒ **Work is completed and shareable**

☒ **Deliverables are produced**

☒ **Documentation is completed**